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0.1 Overview

The NA62 experiment is a fixed target experiment located in the north area of CERN with a high energy particle beam provided by the Super Proton Synchrotron. The main goal of the experiment is to measure the branching fraction of the ultra-rare charged kaon decay $K^+ \to \pi^+\nu\bar{\nu}$ which is of the order of 10^-10 to a 15% precision. Alongside there is a expansive physics program of precision and rare decay measurements as well as searches for forbidden processes and exotic particles.

The experiment employs a decay-in-flight technique to obtain a large number of charged kaon decays, in the order of 10^{13} within the short time frame of a few years. The experimental setup was designed to fulfill all the requirements for the challenging meaurement of $K^+ \to \pi^+\nu\bar{\nu}$, which involves the detection of the K^+ upstream and the detection of the product π^+, with a missing energy-momentum carried away by the undetected neutrino-antinutrino pair. The measurement of the missing mass is used for kinematic rejection of the main background K^+ decays; $K^+ \to \pi^+\pi^0$, $K^+ \to \pi^+\pi^- + \pi^0$ and $K^+ \to \mu^+\nu_\mu$ by selecting two regions shown in figure 1. The missing mass calculation requires precise timing and position measurement of the K^+ and π^+ candidates, provided by beam (section 0.3.2) and secondary (section 0.3.3) particle spectrometers. The measured 3-momenta of the candidates allows for the reconstruction of the 4-momenta of the Kaon P_{K^+} and the pion P_{π^+} which is used in equation 0.1

$$m^2_{miss} = (P_{K^+} - P_{\pi^+})^2 \tag{0.1}$$

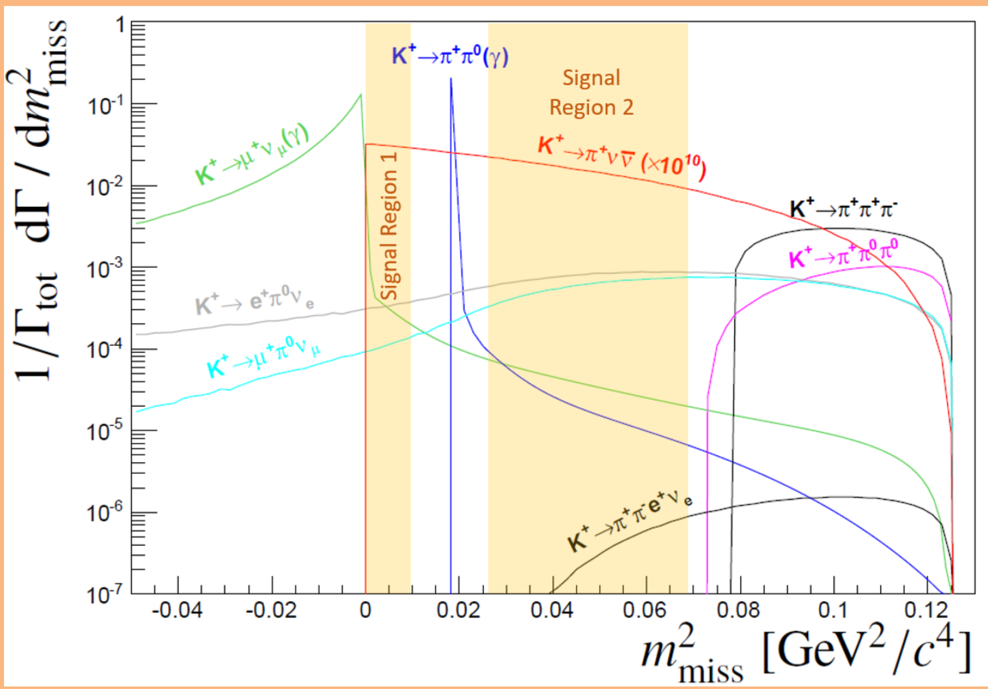

Figure 1: Expected squared missing mass distributions of main kaon decays and $K^+ \to \pi^+\nu\bar{\nu}$ (scaled by 10^{10} for for visability). The solid coloured areas show the signal regions used to reject the domiant kaon decays kinematically.

The rest of this chapter will cover how NA62 fulfills the abovementioned requirements. Section 0.2 will describe the NA62 beam-line, upstream of the NA62 detector. The NA62 detector comprises a number of sub-detectors, which will be described in Section **??**, with the trigger and data acquisition system (TDAQ) following in Section 0.4

0.2 The NA62 Beam Line

The experiment at CERN uses a hadron beam extracted from the SPS with a nominal momentum of 400 GeV/c, directed onto a 400 mm long, 2 mm diameter berylium rod that forms the fixed target (T10) located at 0 m in figure 3. The beam derived from the T10 target, referred to as K12, is a high-intensity unseparated hadron

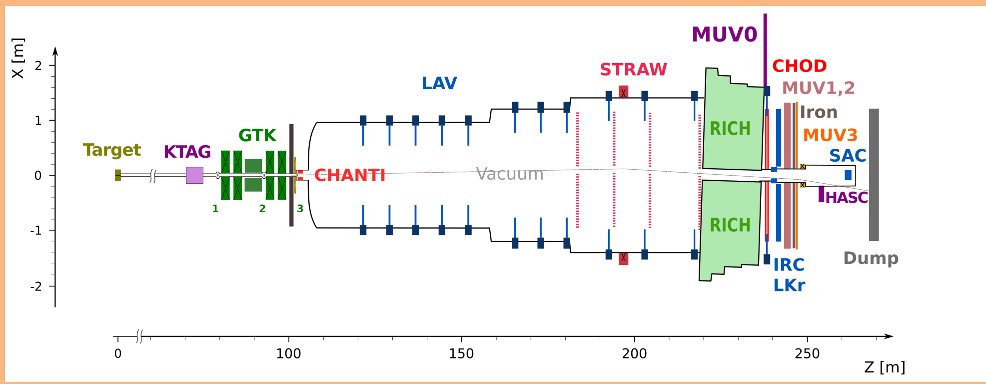

Figure 2: X-Z diagram of NA62 detector setup for Run 1

beam with a nominal momentum of 75 GeV/c. The beam momenta was chosen to maximise the fraction of kaons with respect to the other beam components at the entrance to the detector.

The elements of the K12 beam-line are shown in Figure 3 and comprise a number of components. First are three quadrupole magnets (Q1, Q2 and Q3), used to focus the beam with a large solid angle acceptance of ±2.7 mrad horizontally and ±1.5 mrad vertically at a 75 GeV/c central momentum. Next follows an achromat (A1), which selects a beam of a 75 GeV/c momentum with a 1 % rms momentum. The A1 achromat comprises four dipole magnets that vertically deflect the beam, the first two which deflect the beam 110 mm downwards, while the final two deflect the beam upwards towards the original axis. Between these pairs of magnets, there are a pair of water-cooled beam-dump units (TAX1 and TAX2) with a set of holes to select the required 75 GeV/c momentum beam, while absorbing unwanted secondary beam particles and any remaining primary protons. TAX1&2 can be closed to act as a secondary target for use in dump mode or for safety when access is required downstream. At the focus between the TAX1&2, a radiator consisting of tungston plates with a thickness of up to 5 mm is introduced into the path of beam. The radiator is optmised to reduce positron energy while minimising the loss of energy

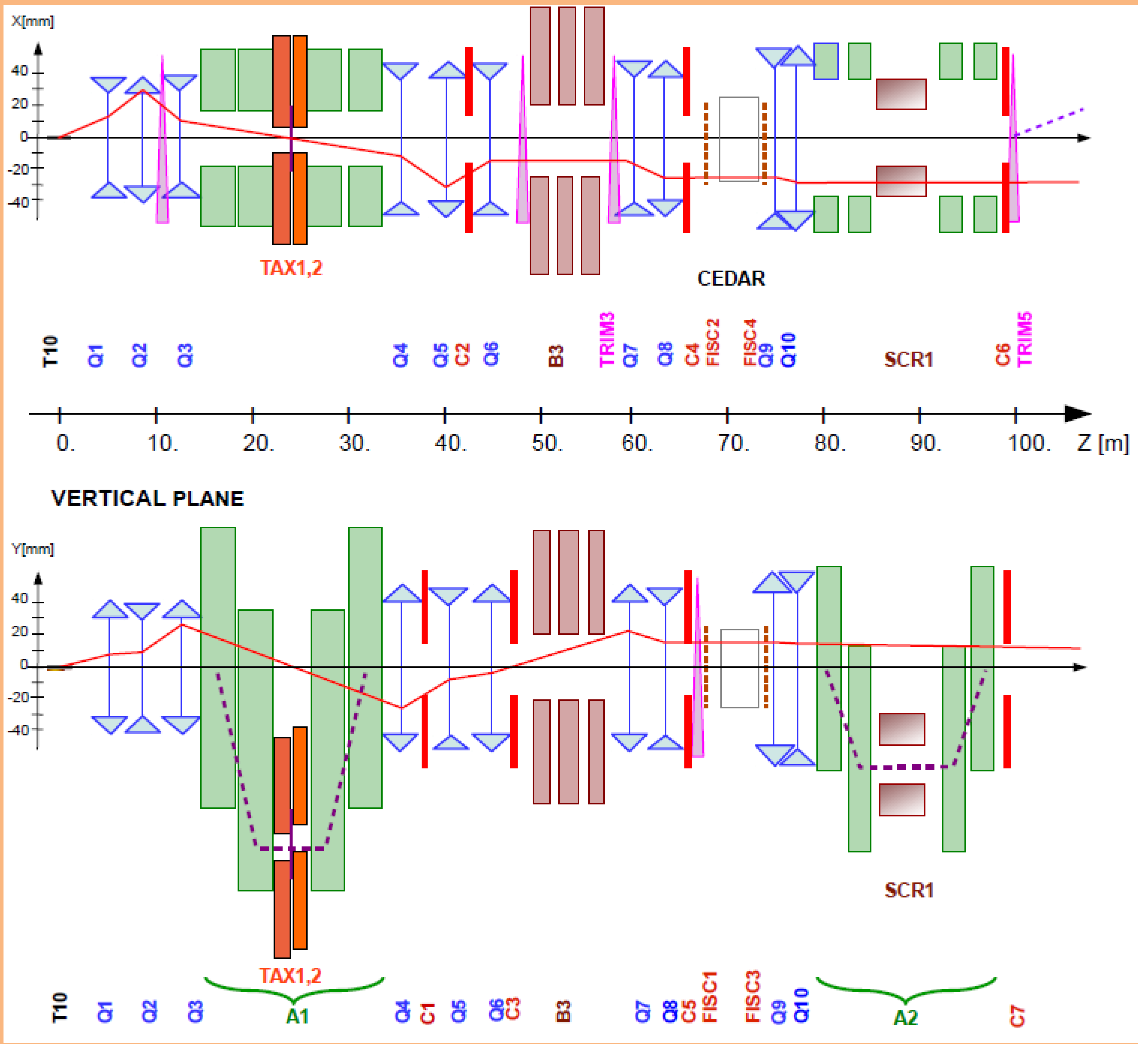

Figure 3: Schematic of K12 beamline from T10 target at 0m to the entence of the NA62 decay region at 102.4m[?].

to hadrons. A triplet of quadrupoles (Q4, Q5, Q6) follows, which refocuses the beam in the vertical dimension and produces a parallel beam in the horizontal dimension. The space between the quadrupoles are occupied by a pair of collimators (C1, C2), which redefines the horizontal and vertical acceptance for the beam. The collimator after the Q6 quadrupole (C3) is employed to absorb positrons that had their momenta degraded by the radiator and again redefines the acceptance in the vertical dimension.

The beam next passes through a 40 mm field-free hole in a set of iron plates, placed between three 2 m dipole magnets (B3). The vertical magnetic field in the iron plates is used to deflect muons of both signs away from the beam. Any deviation in the beam due to stray fields in the beam hole are canceled out by a pair of steering dipoles (TRIM2 and TRIM3). Two quadrupoles (Q7, Q8) follow and ensure that the beam is parallel before the KTAG which employs a differential Cherenkov counter (CEDAR), described in Section 0.3.1, which requires the beam to be parallel. Before the KTAG, a pair of collimators (C4 and C5) are used to absorb particles in the tails of the beam.

For beam monitoring, two pairs of filament scintillator counters (FISC 1, 3 and FISC 2, 4) are installed at either end of the KTAG. Used in coincidence, the beam divergence can be measured and used to tune the divergence to zero.

Following the KTAG are two quadrupoles (Q9 and Q10), used to focus the beam before the beam tracking system (GTk). The GTk comprises four stations made up of silicon pixel detectors described in Section 0.3.2 installed in the beam vacuum. The GTk stations are installed so that the space between GTK1 and GTk3 is occupied by a second achromat (A2), which comprises four dipole magnets that deflect the beam 60 mm vertically, allowing for the beam particle momenta to be measured. The return yokes of the third and fourth magnets and a magnetised iron collimator (SCR1), defocus beam muons which leave the beam between the second and third magnets. A pair of cleaning collimators (C6 and C7) used to absorb beam particles outside of the acceptance are located before the final GTk station (GTk3), which is at the entrance to the decay region ($Z = 102.4\,m$). A steering magnet (TRIM5) is then used to deflect the beam in the horizontal dimension by an angle of +1.2 mrad.

At $Z = 102.4\,m$ begins a large 117 m vacuum pipe (Bluetube), evacuated to a pressure of 10^{-6} mbar by several cryo pumps along its length. The Bluetube is made up of 19 cylindrical sections ranging in diameter shown in Figure 2, from 1.92 m in the first section after GTK3, to 2.4 m in the middle section and then 2.8 m in the region containing the STRAW spectrometer. The first section is defined as the decay region between $Z = 105\,m$ and $Z = 170\,m$. Further downstream, the Bluetube hosts a number of detectors: Eleven LAV stations used in the Photon detection system described in Section 0.3.5 and four STRAW spectrometer chambers coupled with a large aperture dipole magnet (MNP33), providing kinematic measurements of the charged products of kaons decaying in the decay region, described in Section 0.3.3. The MNP33 magnet provides a 270 MeV/c momentum kick in the horizontal dimension, deflecting the 75 GeV/c beam by -3.6 mrad, so that combined with the TRIM5 deflection, the beam passes through the central aperture of the Liquid Krypton calorimeter (LKr). The path of the beam is shown in Figure 4, showing the deflection in the horizontal dimension. After the final STRAW spectrometer chamber (STRAW4), the Bluetube is closed off with a thin aluminum window (except for the hole for the beam to pass through), separating the vacuum from the 17 m long, neon-filled Ring Imaging Cherenkov counter (RICH), designed to distinguish between pions and muons produced in kaon decays and is part of the Particle Identification system (PID) and is described in Section 0.3.4. The beam is transported through the remaining downstream detectors in vacuum and is deflected -13.2 mrad in the negative direction by a dipole magnet (BEND) to sweep the beam away from the Small Angle Calorimeter (SAC) acceptance, located within the vacuum beam pipe. Finally, the beam is absorbed in a beam dump.

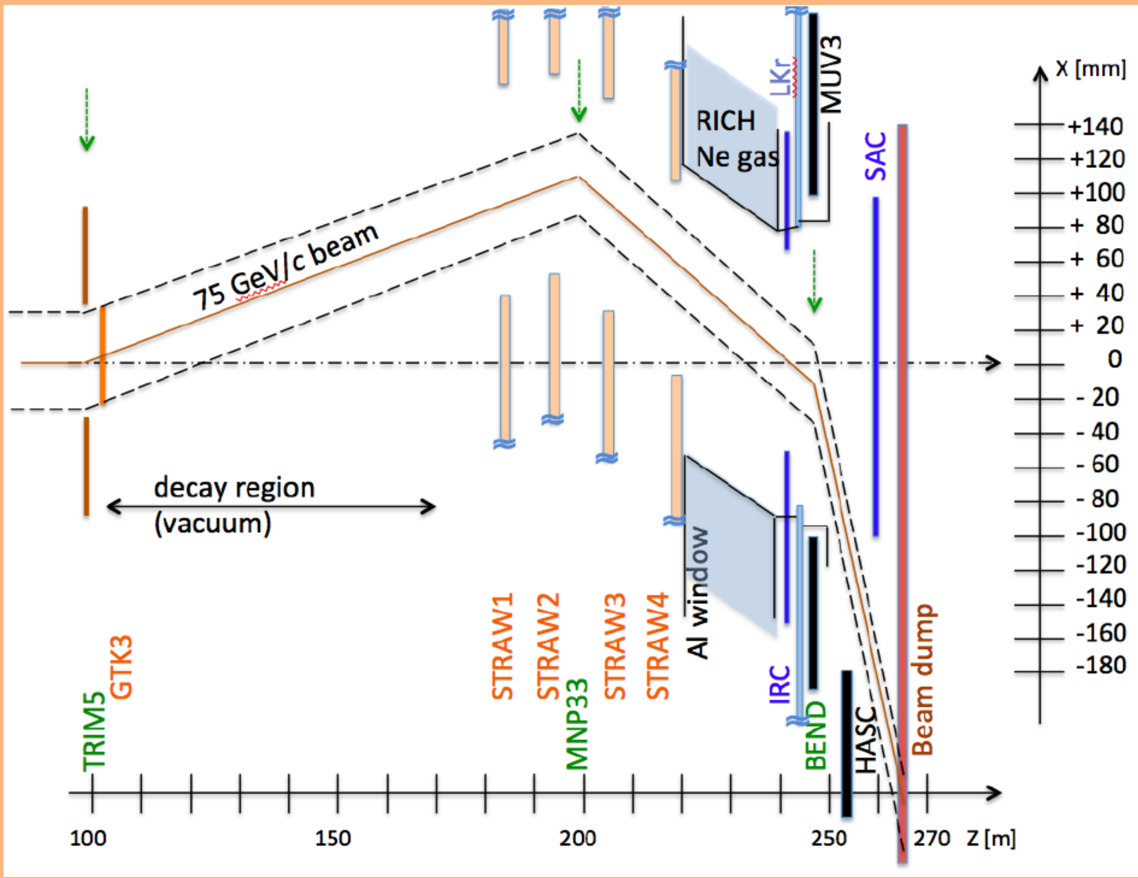

Figure 4: Schematic of K12 beamline after GTK3 [?].

0.3 NA62 Detectors

0.3.1 KTAG

As the NA62 beam is unseparated and kaons only account for 6% of the beam components, the kaons must be identified. Kaon identification is provided by the KTAG and located at $Z = 69.4\,m$. It comprises a CEDAR vessel and a purpose-built photon detection system. Until the NA62 2023 run, the CEDAR vessel used was a CEDAR-W filled with a Nitrogen radiator. CEDARs were developed at CERN in the 1970s [?] for secondary beam particle detection at the super proton synchrotron (SPS). The CEDAR is a differential Cherenkov Counter and exploits the dependence of the beam particle's mass on the properties of the produced light, defined in equation 0.2 where $\beta = \frac{p}{\gamma m}$.

$$cos(\theta) = \frac{1}{\beta n} \tag{0.2}$$

In a beam of set momentum and a radiator with a constant refractive index, the angle of Cherenkov light is only affected by the mass of the beam particle. The pressure is set at 1.7 bar of Nitrogen in the CEDAR-W so that the light from the kaon is focused by a set of lenses and mirrors onto arrays of photomultiplier tubes (PMTs). The light from the other beam components does not reach the PMT arrays. A diagram showing the internal optics of the CEDAR is shown in Figure 5.

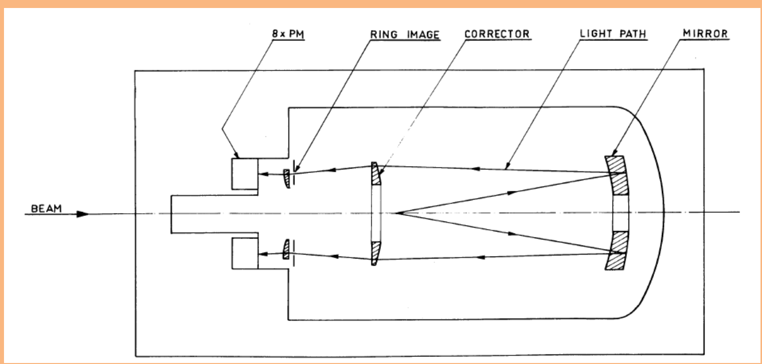

Figure 5: Diagram describing the CEDAR optics [?]

In the original CERN design, the Cherenkov light exiting each CEDAR exit window was detected by an ET-9820QB photomultiplier tube (PMT) placed directly on the exit window, with a particle identified if there was a coincidence of at least six PMTs. However, these PMTs would not fulfill the timing ($\leq$ 100 ps) and rate requirements ($\tilde{4}$5 MHz of kaons) at NA62.

For NA62, a purpose-built photon-detection system was designed, constructed and mounted to the upstream end of the CEDAR Vessel. The photon light from the CEDAR exit windows are reflected radially 90 degrees by eight spherical mirrors onto 8 PMT planes, referred to as sectors. Each PMT plane is equipped with an array of

48 single-anode Hamamatsu PMTs of two types: 32 R9880-110 and 16 R7400. The KTAG photo-detection system allowed the use of at least five sectors for particle identification, resulting in an improved kaon identification efficiency compared to the original setup. A kaon time resolution of 70 ps and a kaon identification efficiency of 95% was achieved with the KTAG and CEDAR-W [?].

In 2023, a new CEDAR with a Hydrogen radiator (CEDAR-H) was installed in the NA62 beamline after design at the Univeristy of Birmingham and testing at CERN. The new CEDAR uses the same basic principles as previous CEDARs but uses Hydrogen as a radiator gas, which required the design of new optical elements in the CEDAR and the photon detection system. A more detailed description of the CEDAR and its optical elements, along with the design, testing and installation of the new CEDAR-H can be found in Section ??.

0.3.2 GigaTracker (GTK)

Figure 6: Image of GTk station [?].

The precise measurement of the beam momentum, time, and position is provided by the GigaTracker (GTk), a beam spectrometer comprising hybrid silicon trackers placed within the vacuum pipe along with an achromat upstream of the decay volume. As particles of differing momenta are deflected by different amounts in a magnetic field, the momentum of a particle can be measured by comparing the positions of the particles in each station. The resolution of the momentum measurement is,

$$\frac{\Delta p}{p} = 0.2\,\% \tag{0.3}$$

with a direction at the exit of the achromat is 16 μrad and a hit time resolution of 200 ps. During run one data-taking period (2016-2018), three stations were employed and are shown in Figure 7. Another station was added (GTK0) for run2 data taking (2021-present) and placed in front of the first GTK station, increasing the tracking efficiency and reducing the backgrounds for the main decay mode. The layout of the GTK is shown in Figure **??**

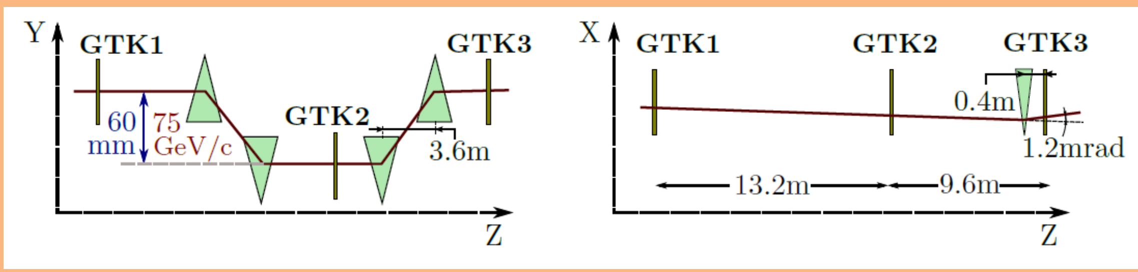

Figure 7: Schematic of GTK stations within the acromat [**?**] for run 1.

Each station consists of 18000 pixels of size $300 \times 300\,\mu m^2$ in a matrix of 200×90, covering a total area of $62.8 \times 27\,mm^2$ and is read out by application-specific integrated circuits (ASIC). The tracker was placed within the beam, so the material has to be minimised to reduce the number of interactions with the beam and improve the angular resolution. The material budget for each station was set at 0.5 %, corresponding to 500 μm of Silicon.

0.3.3 Straw Spectrometer (Straw)

The STRAW Spectrometer provides kinematic measurements of charged decay products produced in K^+ decays within the decay volume. It comprises four chambers and a large aperture dipole magnet (MNP33), providing a momentum resolution of

$$\frac{\sigma(p)}{p} = 0.30\% \oplus 0.005\% \tag{0.4}$$

where p is the track momentum in GeV/c. The chambers are downstream from the decay region, with the first chamber located at $Z = 183\,\mathrm{m}$ and the second 10 m further downstream. The MNP33 magnet is situated downstream of chamber 2 at $Z = 197.6\,\mathrm{m}$ and provides an integrated field of 0.9 Tm. Chambers 3 and 4 are located downstream of the MNP33, $Z = 204\,\mathrm{m}$ and 219 m respectively.

Each chamber comprises four views (X, Y, V, and U) rotated at 0 deg, 90 deg, +45 deg, and −45 deg, each consisting of four layers of straws shown in Figures 8 and 9. The views have a gap near the center, of around 12 cm with no straws, such that when they are combined to form a chamber, an octagonal hole with an apothem of 6 cm is produced for the beam to pass through. As shown in Figure 4, the beam hole for each chamber is not centered to the angle of the beam before and after the MNP33 magnet, with the beam hole parameters defined in Table 1.

Chamber	Beam Hole Size (cm)	Beam Hole Offset (X, Y, Z)
1	6	101.2 mm, 0 mm, 183 m
2	6	114.4 mm, 0 mm, 193 m
MNP33	-	-, -,198 m
3	6	92.4 mm, 0 mm, 204 m
4	6	52.8 mm, 0 mm, 218 m

Table 1: Straw spectrometer chamber beamhole positions

The straws are 2.1 m long drift tubes that are 9.82 mm in diameter, made from 36 μm thick polyethylene terephthalate (PET), coated on the inside with 50,nm of copper and 20 nm of gold. The Straws are filled with a mixture of 70% Argon and 30% CO_2 at atmospheric pressure, and have a central gold-plated tungsten anode wire. The straws act like proportional counters, when a charged particle passes through the gas, it is ionised, creating electron-ion pairs that drift to the opposite electrodes inducing a charge on the anode wire. When the electron approaches the anode wire, the electric field strength increases dramatically, producing Townsend avalanches. This provides a large multiplication of the signal, producing a charge pulse, which can be read out.

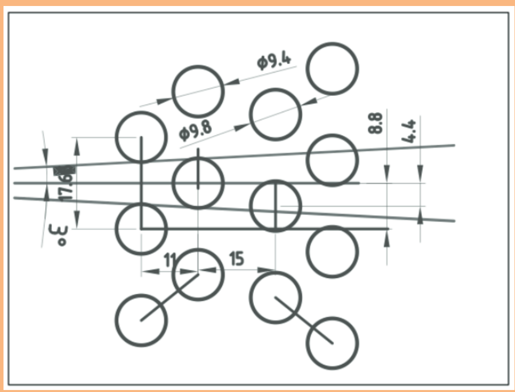

Figure 8: Diagram of straw placement in each view [?]

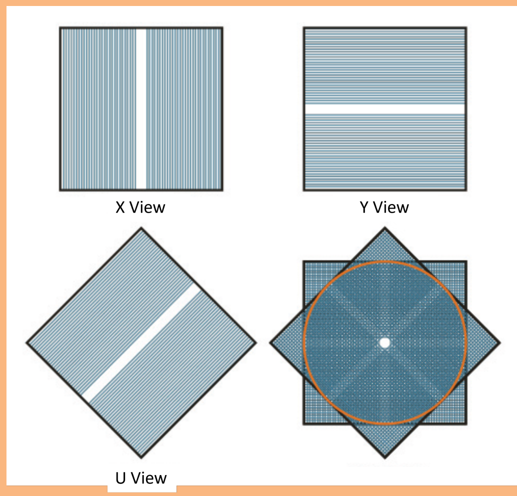

Figure 9: Diapgram showing placement of views in a straw chamber. Edited from [?]

0.3.4 Ring Imaging Cherenkov counter (RICH)

The Ring Imaging Cherenkov counter (RICH) was designed to separate secondary pion and muons within a momentum range of 15 - 35 GeV/c. The ability to separate the secondary particles is required as the largest background to $K^+ \to \pi^+\nu\bar{\nu}$ comes from the $K^+ \to \mu^+\nu_\mu$, which has a BR 10 orders of magnitude greater. The majority of the suppression comes from kinematic selections and the response difference between pions and muons in the calorimeters, however, another factor of 100 is required in the momentum range 15 - 35 GeV/c was required, which the RICH satisfied.

The RICH is a 17.5 m long cylindrical vessel comprising four steel sections with diameters gradually decreasing from 4.2 m at the upstream section to accommodate the PMT flanges to 3.2 m in the downstream end. The radiator vessel is separated from the vacuum decay tube by an aluminum window of 2 mm thickness and radius of

1.1 m at the upstream end, allowing minimal material in the path of particles entering the detector. A lightweight aluminum pipe connected to the vacuum tube passes through the vessel, allowing the beam to pass through in a vacuum and continues through the 4 mm thick, 1.4 m radius aluminum exit window. The vessel contains Neon gas at a pressure of 900 mbar with a refractive index of $n = 1 + 62.8 \times 10^{-6}$ at a wavelength of 300 μm, corresponding to a Cherenkov threshold for charged pion of $12.5 GeV/c$.

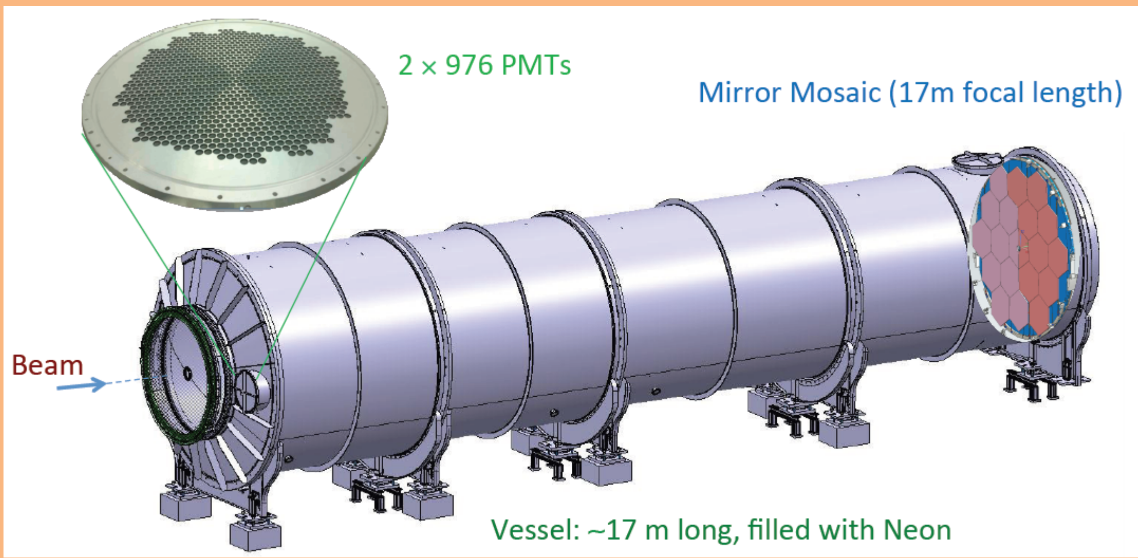

Figure 10: Schematic of RICH detector, showing the beam entering the upstream end on the left. A zoomed-in image of one of the two PMT flanges shows the layout of the PMT arrays. The mirrors are also shown at the dowstream end of the vessel [?]

A mosaic of 20 spherical mirrors (18 hexagonal and 2 semi-hexagonal) with a focal length of 17 m are located at the downstream end of the vessel and is shown in Figure 10. This mirror system is used to reflect and focus the Cherenkov light toward two PMT arrays located at the upstream end of the vessel. The two PMT arrays are located on either side of the entrance window (see Figure 10), each consist of 976 Hamamatsu R7400-U03 PMTs. Each PMT has a 8 mm active area and are packed into a hexagonal lattice, each with a Winston cone directing light onto the active area.

The particles are identified (selected or rejected) using a cut on the particle's mass, calculated from the particle velocity using the reconstructed Cherenkov ring radius shown in Figure 11 and the momentum of the particle measured by the spectrometer. In Figure 11, it is clear that separating pions and muons within the desired momentum range is possible.

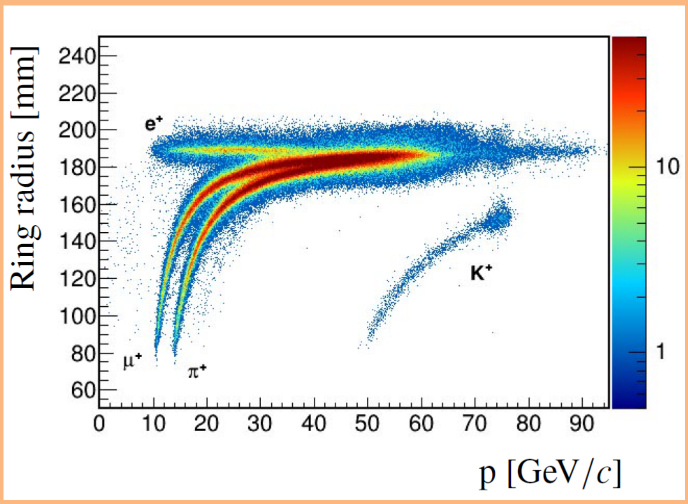

Figure 11: Cherenkov ring radius as a function of the particle momentum [?]

0.3.5 Photon Detection System

0.4 Triger and Data Acquisition System (TDAQ)